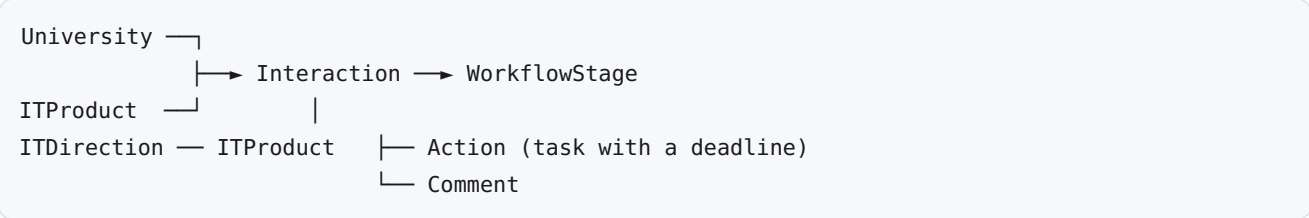
Data schema

English version of ../../database/schema.md. The full ER model is in architecture/er-model.md (plus er.mmd, er-model.pdf). This page covers data ownership and the rules.

Business model



University, ITProduct, ITDirection are reference data. Interaction is the main entity: the "university + product" pair moving through the stages.

Data ownership

Domain	Writer	Reader
Reference data (universities, products, directions)	api.directories, admin only	all modules, reports
Workflow stages	api.stages, admin only	interactions, reports
Interactions, tasks, comments	api.interactions	reports, notifications, audit
Audit	middleware.audit	the administrator
Files	services.file_storage (S3/MinIO), references in the database	export, PDF guides

Rules

1. **The limit "at most 2 active interactions per university"** is enforced by the application. There is no schema constraint — a known limitation (see the audit).
2. **Reference data is never deleted physically** while it is referenced: deactivate first.
3. **Cyrillic in exports** is a cross-cutting requirement: the Excel export is covered by a test.
4. **Production is PostgreSQL 16**, SQLite is acceptable locally; SQLite-only changes are not considered sufficient.

Verification

- Model/migration consistency — make alembic-check.
- Installation from scratch and re-application — portability.yml (alembic-fresh, alembic-upgrade-path).
- Restoring a backup — portability.yml (backup-restore).